PCD Assignment 03

Image Segmentation and Morphological Processing

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1. Introduction

Image segmentation is one of the techniques used in digital image processing to separate objects from their background. It helps identify certain parts of an image so they can be processed or analyzed more easily. One common segmentation method is thresholding, which separates pixels based on their intensity values. However, the result of segmentation is not always perfect because an image may contain noise, small gaps, or objects that are close to each other.

In this assignment, I used Otsu's thresholding to segment an image of rice grains. This method was chosen because it can automatically determine a threshold value based on the distribution of pixel intensities. After segmentation, I applied four morphological operations, which are erosion, dilation, opening, and closing. These operations were used to observe how the shapes and boundaries of the segmented objects change.

The main purpose of this experiment was to compare the segmentation result before and after morphological processing. All segmentation and morphological algorithms were implemented manually without using built-in image processing functions.

2. Methodology

The experiment was conducted using Python in Google Colab. The input image was ricegrain.jpg, which contains rice grains that need to be separated from the background. Before applying segmentation, the image was converted into grayscale and resized to 256 × 256 pixels. This was done to simplify the process and reduce the computation time, since the algorithms were implemented using loops.

The first step was implementing Otsu's thresholding. This method works by calculating the pixel intensity histogram and selecting a threshold that maximizes the variance between the foreground and background classes. In this experiment, the histogram and threshold calculation were implemented manually. After finding the threshold, each pixel was classified into either foreground or background to produce a binary image. White pixels represented the detected foreground, while black pixels represented the background.

The next step was applying morphological processing using a 3 × 3 square structuring element. Four operations were implemented. Erosion reduces the size of foreground regions by removing pixels around their boundaries, while dilation expands the foreground regions by adding pixels around the objects. Opening is performed by applying erosion followed by dilation, which can remove small foreground noise. Closing uses dilation followed by erosion, which can fill small holes and connect nearby foreground regions.

Each operation was implemented manually using nested loops to examine neighboring pixels. The results of erosion, dilation, opening, and closing were displayed separately to compare their effects. For the final processing result, opening was applied first, followed by closing.

3. Results and Discussion

Based on the experiment, Otsu's thresholding produced a threshold value of 101. This value was used to convert the grayscale rice grain image into a binary image. The initial segmentation was able to separate many rice grains from the background. However, the result still contained some small gaps, and several grains were already close to or touching each other.

The morphological operations produced different changes to the segmented image. Erosion made the white regions smaller because it removed pixels from the boundaries of the rice grains. Some small foreground details were also removed. In contrast, dilation made the white regions larger and caused some neighboring grains to become more connected.

Opening helped remove small foreground details while keeping the larger regions relatively similar to their original shapes. Closing filled some of the small gaps and made the foreground regions more continuous. When opening and closing were combined, the final segmentation appeared smoother in some areas, although a few rice grains became connected.

The comparison also showed differences in the number of foreground pixels. Before morphological processing, the segmented image contained 27,066 foreground pixels. After applying opening and closing, the number increased to 28,549 pixels. A total of 2,687 pixels changed their classification between the two results.

The increase in foreground pixels shows that the final processing added more foreground pixels than it removed. This can happen because closing fills gaps and expands the connected foreground regions. However, having more foreground pixels does not always mean the segmentation is more accurate. In this experiment, some rice grains became harder to distinguish individually because their boundaries were connected after processing.

From the comparison, I observed that morphological processing is useful for modifying the segmentation result, especially for removing small details and filling gaps. However, the results also depend on the structuring element and the distance between objects. Since rice grains can be

close together, morphological operations need to be applied carefully to avoid merging objects that should remain separate.

4. Conclusion

In this assignment, I implemented Otsu's thresholding and four morphological operations manually without using built-in image processing functions. Otsu's thresholding successfully separated the rice grains from the background using a threshold value of 101.

The comparison showed that morphological processing changed the shapes and boundaries of the segmented objects. Opening helped remove small foreground details, while closing filled gaps and made some regions more continuous. However, the processing also connected some neighboring rice grains.

Overall, this experiment helped me understand how segmentation and morphological processing work together. Morphological operations can improve certain parts of a segmented image, but they do not always produce a better result for every object. The appropriate operation depends on the image and what we want to achieve from the segmentation.